DecisionRL: Enhancing Decision Calibration through Reinforcement Learning

Reinforcement Learning Course Project Proposal

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Research Question

As large language models evolve from question answerers to decision makers, efficient and reliable decision-making becomes increasingly important. While System 2-style models emphasize deliberate, multi-step reasoning, System 1-style models aim to provide fast, intuitive judgments. Jev introduces a complementary paradigm of structured probabilistic decisions, producing explicit probability distributions over candidate actions instead of lengthy reasoning trajectories. Inspired by this paradigm, our project explores how **Reinforcement Learning for Calibrated Decisions (RLCD)** can enhance decision accuracy, probability calibration, and reliability through a reproducible training framework.

Related Work

Jev introduces System 1-style probabilistic decision-making and RLCD, while **Microsoft-Decision-1** demonstrates the potential of efficient decision models. However, their proprietary implementations and limited disclosure of training methodologies restrict reproducibility. **Clef** provides open model weights but does not offer a fully reproducible training pipeline. Meanwhile, **Laya** and **Kev** provide open-source implementations of typed-decision models. Laya introduces a practical RLCD training approach, whereas Kev primarily adopts supervised learning and post-hoc calibration. However, RLCD optimization remains relatively underexplored, particularly in reward design and policy optimization. Our project aims to develop a reproducible RLCD training framework and explore improvements in reinforcement learning strategies.

Planned Approach

We plan to develop **DecisionRL** to explore reinforcement learning strategies for reliable and calibrated decision-making. We will consider **ModernBERT** and **EuroBERT** as pretrained encoder backbones.

- **Data Construction and Curation.** We will use all official `typed-decisions` training examples and collect additional open-source decision datasets across multiple domains. Our data preparation will cover two types of supervision: hard labels for SFT and probabilistic calibration labels for RLCD. Since reliable probabilistic labels are scarce, we will synthesize diverse decision scenarios and construct calibration labels from teacher-model logit distributions or aggregated predictions through repeated sampling and voting.
- **Model Training and RLCD Optimization.** We will adopt a two-stage training framework. First, we will perform supervised fine-tuning (SFT) on high-quality decision datasets to establish general decision-making capabilities. We will then apply RLCD to optimize decision probabilities using verified probabilistic supervision. We also aim to improve Laya's original RLCD algorithm by exploring enhanced reinforcement learning optimization strategies for more reliable and calibrated decisions.

Evaluation Plan

We will compare the original pretrained backbones, released Laya, SFT, SFT + RLCD, and DecisionRL on the held-out test set, and we will report **Accuracy**, **Macro F1 Score**, **Expected Calibration Error (ECE)**, and **Brier Score** to measure decision correctness, class-balanced performance, confidence calibration, and probability quality, respectively.

Required Resources and Division of Responsibilities

Resources: No additional equipment or specialized resources are required.

Preliminary Division of Responsibilities: Both team members will jointly participate in project discussions, research planning, and RLCD methodological design. Yanyun LIU will primarily contribute to data collection, synthesis, and quality control, while Taihua Chen will focus more on model and algorithm optimization.